Inverse of Matrix

Inverse of Matrix:

- If A is a square matrix, and if a matrix B of the same size can be found such that $AB = BA = I$, then A is said to be ***invertible*** (or ***nonsingular***) and B is called an ***inverse*** of A. If no such matrix B can be found, then A is said to be ***singular***.

[**N.B: Singular matrix & non-Singular matrix:** A matrix B is said to be singular if and only if $\mathbf{det(}\boldsymbol{B}\mathbf{) = 0}$, otherwise it called be non-singular, i.e., $\mathbf{det(}\boldsymbol{B}\mathbf{) \neq 0}$.]

$$\begin{bmatrix} 2 & 3 \\ 4 & 6 \end{bmatrix} \qquad\qquad \begin{bmatrix} 2 & -3 \\ 4 & 6 \end{bmatrix}$$

Singular Matrix **Non-Singular Matrix**

- If A is a non-singular square matrix, there is an existence of n x n matrix A^{-1}, which is called the **inverse matrix** of A such that it satisfies the property: $AA^{-1} = A^{-1}A = I$, where I is the Identity matrix.

Formula for finding inverse of a Matrix:

Let A is a nonsingular matrix, then the inverse of A is defined and denoted by

$$A^{-1} = \frac{1}{|A|}\mathrm{adj}A \quad \text{or} \quad A^{-1} = \frac{1}{\det(A)}\mathrm{adj}A$$

Where, adjA is the adjoint matrix of A.

Adjoint matrix: The adjoint of a square matrix $A = [a_{ij}]_{n\times n}$ is defined as the transpose of the matrix $[C_{ij}]_{n\times n}$, where C_{ij} is the cofactor of the element a_{ij}. In other words, the transpose of a cofactor matrix of the square matrix is called the **adjoint of the matrix**. The adjoint of the matrix A is denoted by adj A. This is also known as adjugate matrix or adjunct matrix.

How to Calculate the Adjoint of a Matrix:

Let A be the matrix of order n x n,

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \cdots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix}$$

Now the adjoint of this matrix is:

$$adjA = Transpose\ of \begin{bmatrix} C_{11} & C_{12} & \cdots & C_{1n} \\ C_{21} & C_{22} & \cdots & C_{2n} \\ \vdots & \vdots & \cdots & \vdots \\ C_{n1} & C_{n2} & \cdots & C_{nn} \end{bmatrix}$$

Here, C_{11}, C_{12},…, C_{21}, C_{22},…., C_{nn} are the cofactors of the elements a_{11}, a_{12},…, a_{21}, a_{22},….,a_{nn} respectively.

Method of Finding inverse of a matrix:

Method 1: For 2 x 2 Matrix.

Let A be the matrix of order 2 x 2,

$$\mathrm{A} = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

The inverse of a matrix is found using the following formula:

$$\mathrm{A}^{-1} = \frac{1}{ad-bc}\begin{bmatrix} d & -b \\ -c & a \end{bmatrix}.$$

Example: Find the inverse of the matrix $\mathrm{A} = \begin{bmatrix} 1 & 2 \\ 1 & 3 \end{bmatrix}$.

Solution: The inverse of the matrix A is,

$$\mathrm{A}^{-1} = \frac{1}{1\times3-1\times2}\begin{bmatrix} 3 & -2 \\ -1 & 1 \end{bmatrix} = \begin{bmatrix} 3 & -2 \\ -1 & 1 \end{bmatrix}.$$

Method 2: For n x n Matrix.

One of the most important methods of finding the matrix inverse involves finding the minors and cofactors of elements of the given matrix. Observe the below steps to understand this method clearly.

- The inverse matrix is also found using the following equation:

$$A^{-1} = \frac{1}{|A|}\text{adj}A \quad \text{or} \quad A^{-1} = \frac{1}{\det(A)}\text{adj}A$$

where *adj(A)* refers to the adjoint of a matrix A, *det(A)* refers to the determinant of a matrix A.

A. The adjoint of a matrix A or adj(A) can be found using the following method.

In order to find the adjoint of a matrix A first, find the cofactor matrix of a given matrix and then take the transpose of a cofactor matrix.

- The cofactor of a matrix can be obtained as $C_{ij} = (-1)^{i+j} \det(M_{ij})$.

Here, M_{ij} refers to the $(i,j)^{th}$ minor matrix after removing the *i*th row and the *j*th column.

Example: Find the inverse of the matrix $A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 2 & 9 \end{bmatrix}$.

Solution: Given matrix,

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 2 & 9 \end{bmatrix}$$

Determinant of the matrix is,

$$\det(A) = \begin{vmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 2 & 9 \end{vmatrix} = 1(45-12) - 2(36-42) + 3(8-35) = 33 + 12 - 81 = -36$$

which is not equal to zero, thus this matrix is invertible.

Let us find the cofactor of the given matrix as given below,

$$C_{11} = \begin{vmatrix} 5 & 6 \\ 2 & 9 \end{vmatrix} = 45 - 12 = 33 \quad C_{12} = -\begin{vmatrix} 4 & 6 \\ 7 & 9 \end{vmatrix} = -(36 - 42) = 6 \quad C_{13} = \begin{vmatrix} 4 & 5 \\ 7 & 2 \end{vmatrix} = 8 - 35 = -27$$

$$C_{21} = -\begin{vmatrix} 2 & 3 \\ 2 & 9 \end{vmatrix} = -(18 - 6) = -12 \quad C_{22} = \begin{vmatrix} 1 & 3 \\ 7 & 9 \end{vmatrix} = 9 - 21 = -12 \quad C_{23} = -\begin{vmatrix} 1 & 2 \\ 7 & 2 \end{vmatrix} = -(2 - 14) = 12$$

$$C_{31} = \begin{vmatrix} 2 & 3 \\ 5 & 6 \end{vmatrix} = 12 - 15 = -3 \quad C_{32} = -\begin{vmatrix} 1 & 3 \\ 4 & 6 \end{vmatrix} = -(6 - 12) = 6 \quad C_{33} = \begin{vmatrix} 1 & 2 \\ 4 & 5 \end{vmatrix} = 5 - 8 = -3$$

The cofactor matrix is,

$$\begin{bmatrix} 33 & 6 & -27 \\ -12 & -12 & 12 \\ -3 & 6 & -3 \end{bmatrix}$$

Now the adjoint of the matrix A, i.e., adj(A) is the transpose of the cofactor matrix, we get

$$adj(\mathrm{A}) = \begin{bmatrix} 33 & 6 & -27 \\ -12 & -12 & 12 \\ -3 & 6 & -3 \end{bmatrix}^T = \begin{bmatrix} 33 & -12 & -3 \\ 6 & -12 & 6 \\ -27 & 12 & -3 \end{bmatrix}$$

Hence, the inverse of the given matrix is,

$$\mathrm{A}^{-1} = \frac{1}{\det(\mathrm{A})}\mathrm{adjA} = \frac{1}{-36}\begin{bmatrix} 33 & -12 & -3 \\ 6 & -12 & 6 \\ -27 & 12 & -3 \end{bmatrix} = \begin{bmatrix} -\frac{11}{12} & \frac{1}{3} & \frac{1}{12} \\ -\frac{1}{6} & \frac{1}{3} & -\frac{1}{6} \\ \frac{3}{4} & -\frac{1}{3} & \frac{1}{12} \end{bmatrix} or \frac{1}{12}\begin{bmatrix} -11 & 4 & 1 \\ -2 & 4 & -2 \\ 9 & -4 & 1 \end{bmatrix}$$

Properties of Inverse matrix:

- If A is nonsingular, then $\left(\mathrm{A}^{-1}\right)^{-1} = \mathrm{A}$
- If A and B both are nonsingular matrices, then AB is nonsingular. Thus, $(\mathrm{AB})^{-1} = \mathrm{B}^{-1}\mathrm{A}^{-1}$
- If A is nonsingular, then $\left(\mathrm{A}^{T}\right)^{-1} = \left(\mathrm{A}^{-1}\right)^{T}$

- If A is any matrix and A^{-1} is its inverse, then $AA^{-1} = A^{-1}A = I_n$, where n is the order of matrices.

Exercise

❖ Find the inverse of the following Matrices,

a. $A = \begin{bmatrix} 1 & 2 & 3 \\ 3 & -2 & 1 \\ 4 & 1 & 1 \end{bmatrix}$ b. $B = \begin{bmatrix} 0 & 1 & 2 \\ 1 & 2 & 3 \\ 3 & 1 & 1 \end{bmatrix}$ c. $C = \begin{bmatrix} 0 & 1 & 2 \\ 1 & 2 & 3 \\ 3 & 1 & 1 \end{bmatrix}$

d. $D = \begin{bmatrix} -1 & 2 & 0 \\ 2 & 1 & 0 \\ 4 & -2 & 0 \end{bmatrix}$ e. $E = \begin{bmatrix} 1/5 & 1/2 & -1/3 \\ 2/5 & -5 & -2/3 \\ 3 & 3/2 & 1 \end{bmatrix}$ f. $F = \begin{bmatrix} -1 & 2 & -3 \\ 2 & 1 & 0 \\ 4 & -2 & 5 \end{bmatrix}$

g. $G = \begin{bmatrix} -1 & 1/2 & 0 \\ 0 & 2 & -1/2 \\ 1 & 0 & 3 \end{bmatrix}$ h. $H = \begin{bmatrix} 3/7 & -5/7 & 1/7 \\ 1/7 & -4/7 & -2/7 \\ 0 & -1 & 0 \end{bmatrix}$ i. $I = \begin{bmatrix} -2 & 3 \\ 4 & -5 \end{bmatrix}$